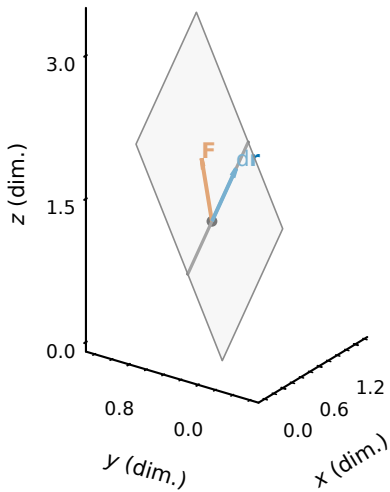
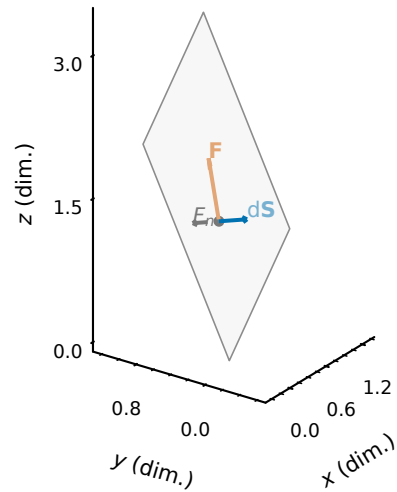
A



$\mathrm{d}s = \sqrt{2}\,\mathrm{d}u$; $\mathbf{F} \cdot \mathrm{d}\mathbf{r} = 2\mathrm{d}u$

B



$\mathrm{d}S = \sqrt{6}\,\mathrm{d}u\,\mathrm{d}v$; $\mathbf{F} \cdot \mathrm{d}\mathbf{S} = -2\mathrm{d}u\,\mathrm{d}v$